

Mobile Network

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2025

Introduction : Evolution

Ver 1.0

Contenu

Planning et conditions d'examen

Réseaux et cellules

Rappel de la théorie de l'information

Les contraintes économiques et techniques

Affaiblissements et Ondes

Evolution des réseaux mobiles

LTE 4G

Réseau 4G

Mobile subscriber : User Equipment

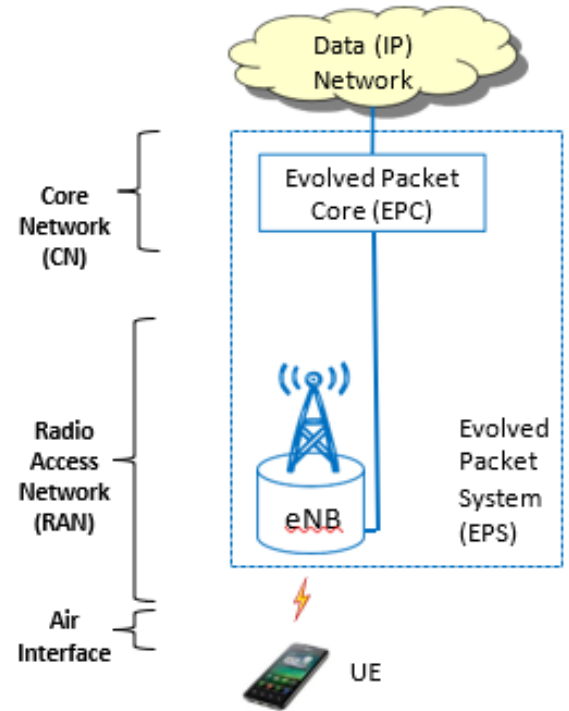
Interface Air

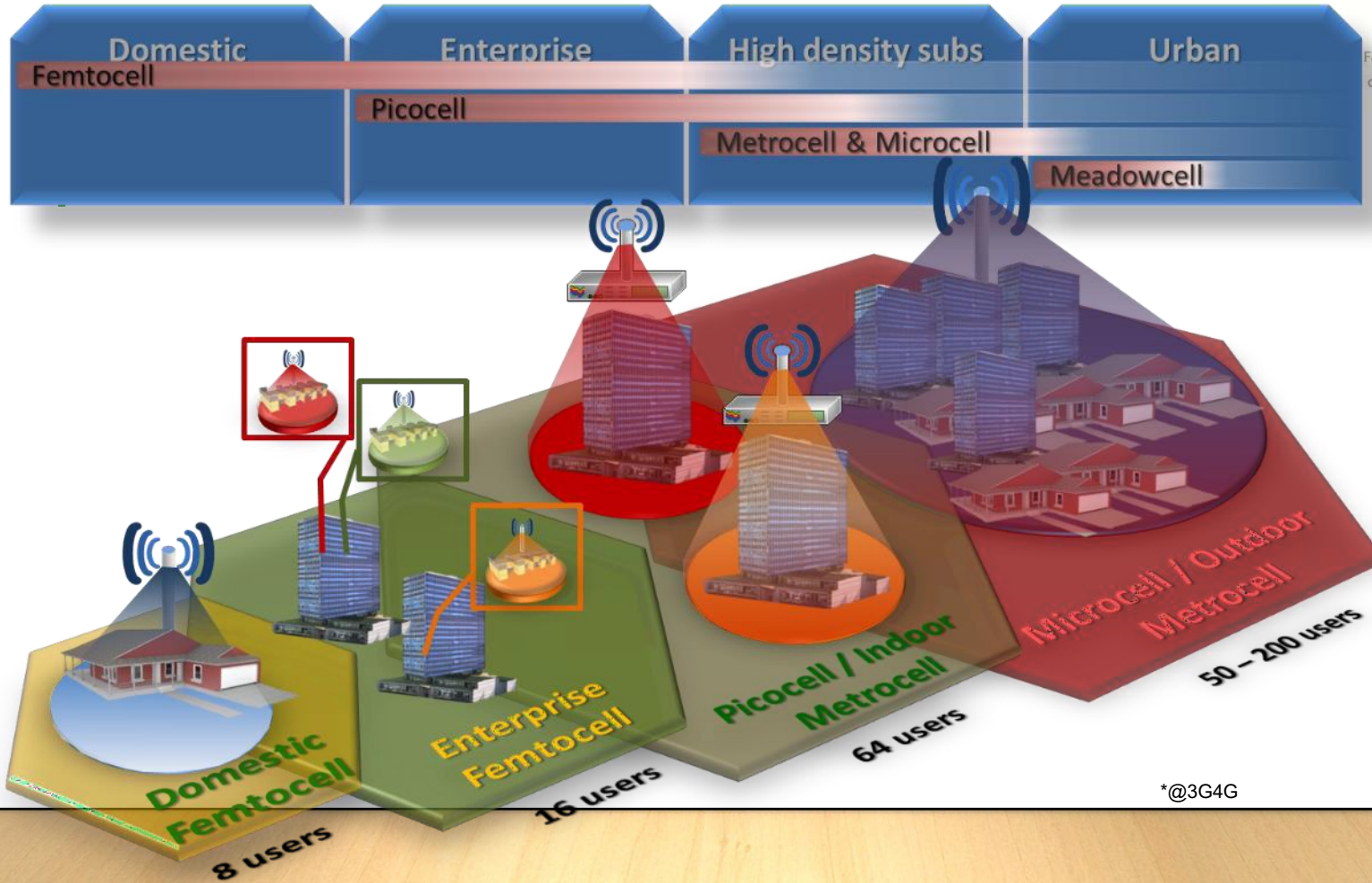
Base station : e NodeB

Contrôleur de station de base

Mobile switching Center : Core Network

Les fréquences ont évolué





Les antennes et mâts



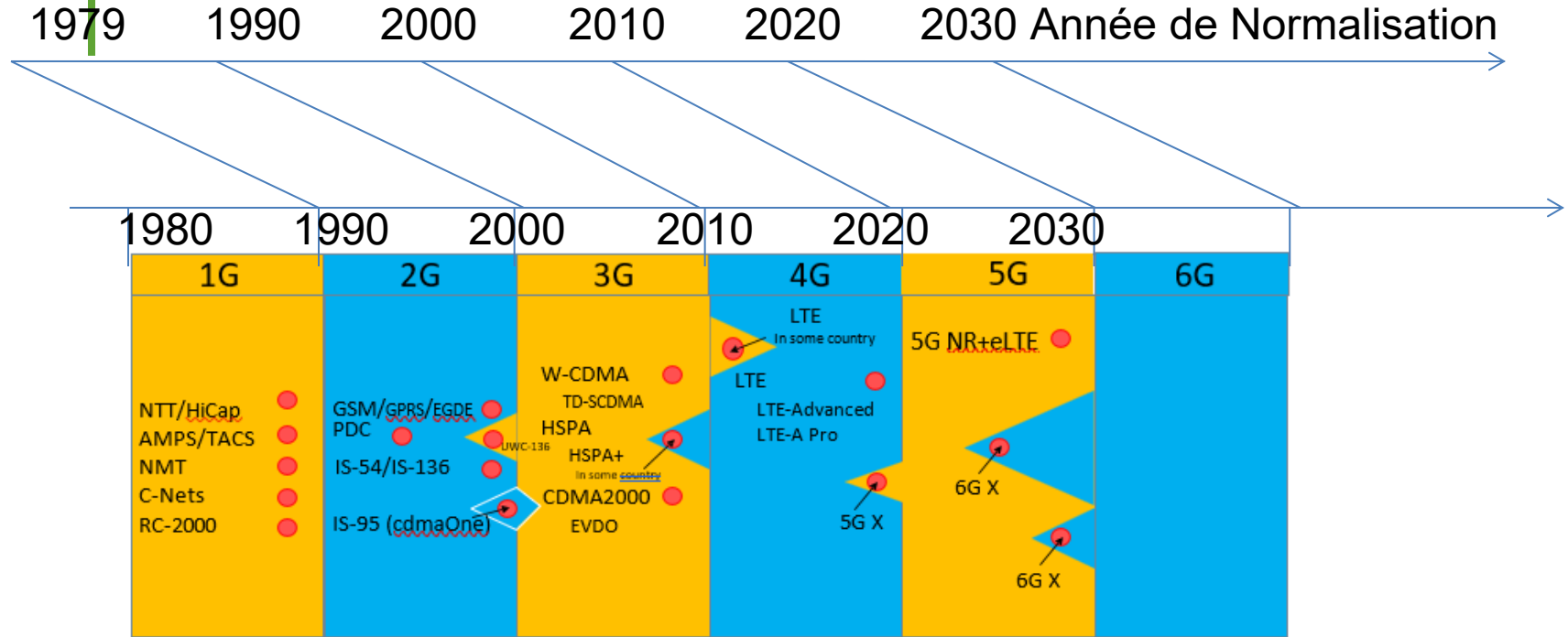
Théorie de l'information

- Claude Shannon (1940) : Mathématicien/Ingénieur électricien
- **Théorème 1** : On peut compresser jusqu'à l'**Entropie**
- **Théorème 2** : Si le débit est inférieur à la **capacité du canal**, le taux d'erreur peut tendre vers zéro moyennant un code correcteur
- **Théorème 3** : La sécurité parfaite est possible avec deux conditions
 - a. La taille de la clef
 - b. La clef doit satisfaire une condition

Entropie : la moyenne de la quantité d'information

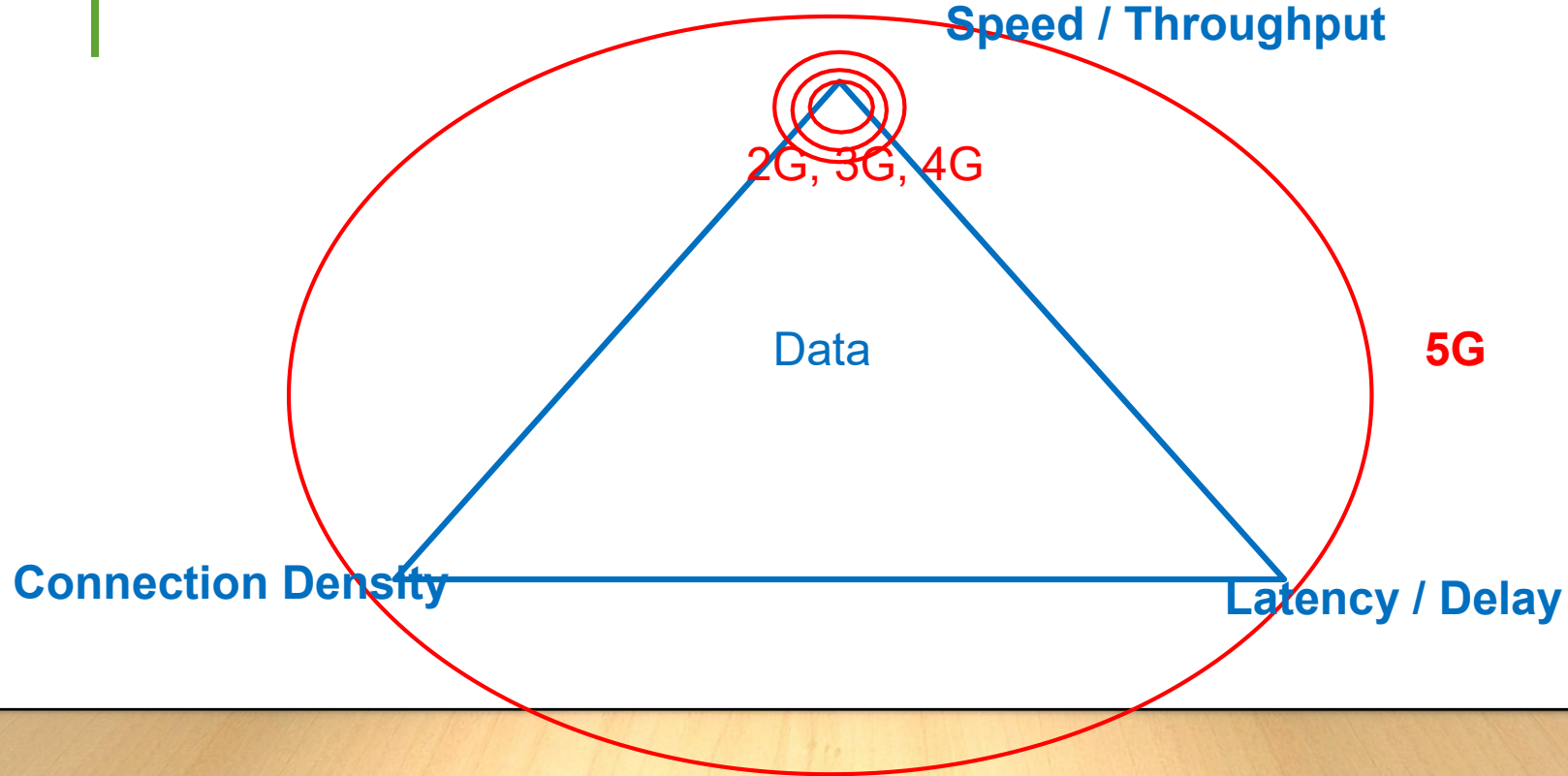
Capacité $C = B \log_2(1 + \text{SNR})$

Evolution



Les normes, la technologie et le marketing s'affrontent en permanence lorsqu'il s'agit d'attribuer des noms aux différentes générations. En général, c'est le marketing qui l'emporte !

Les évolutions



Les contraintes Economiques

Time to Market

Economy of Scale

Economy of Scope

Energy optimization

Autonomy

User Centric

User Experience

Ubiquitous

Contraintes techniques

Couverture globale

Convergence IP

Convergence Fixe_Mobile

Zero_Trust Security

Cross-Layring Security

Couverture Globale

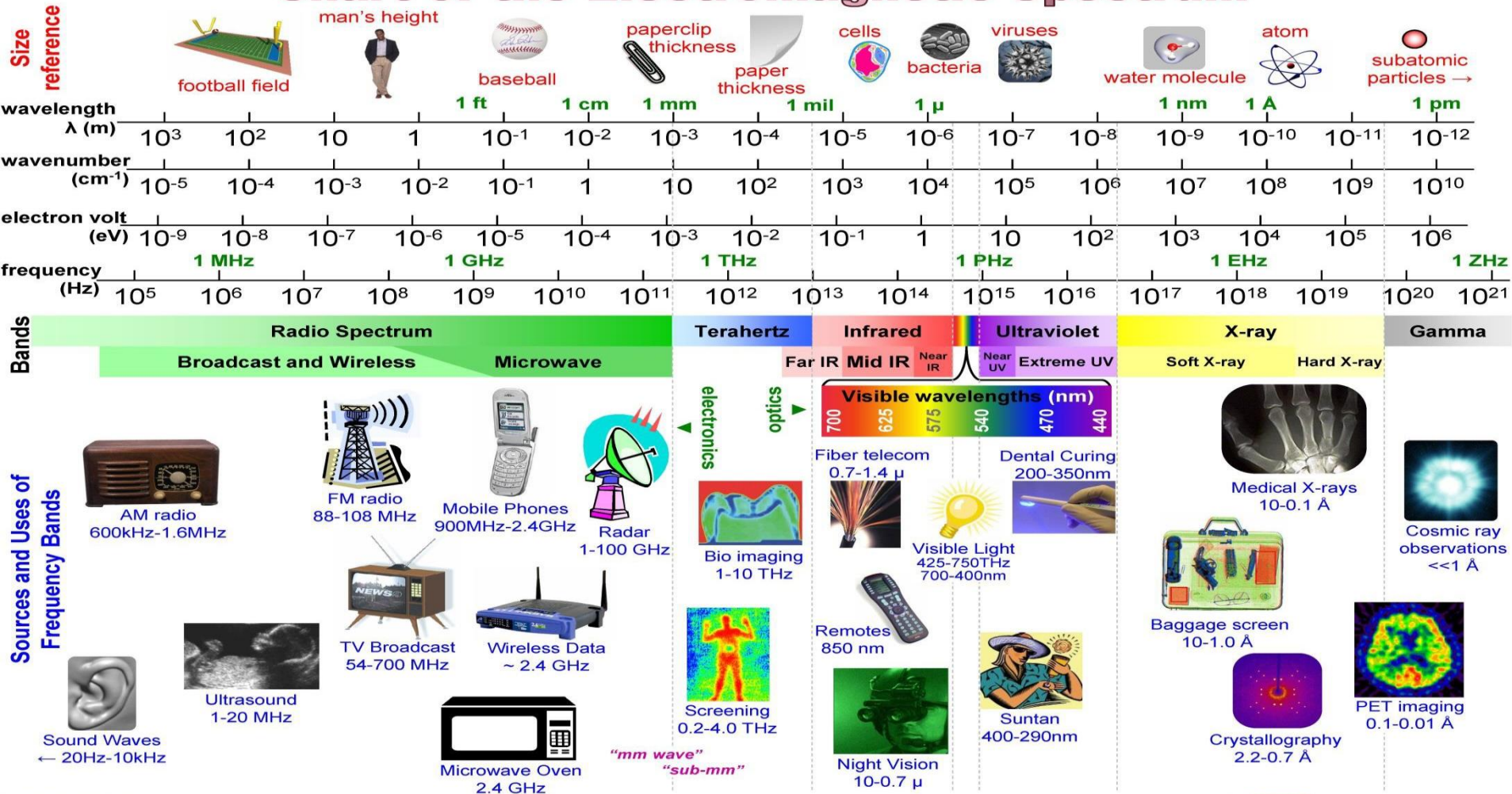
Latence et Gigue

Débit

SDN

Réseaux Privés Open-RAN

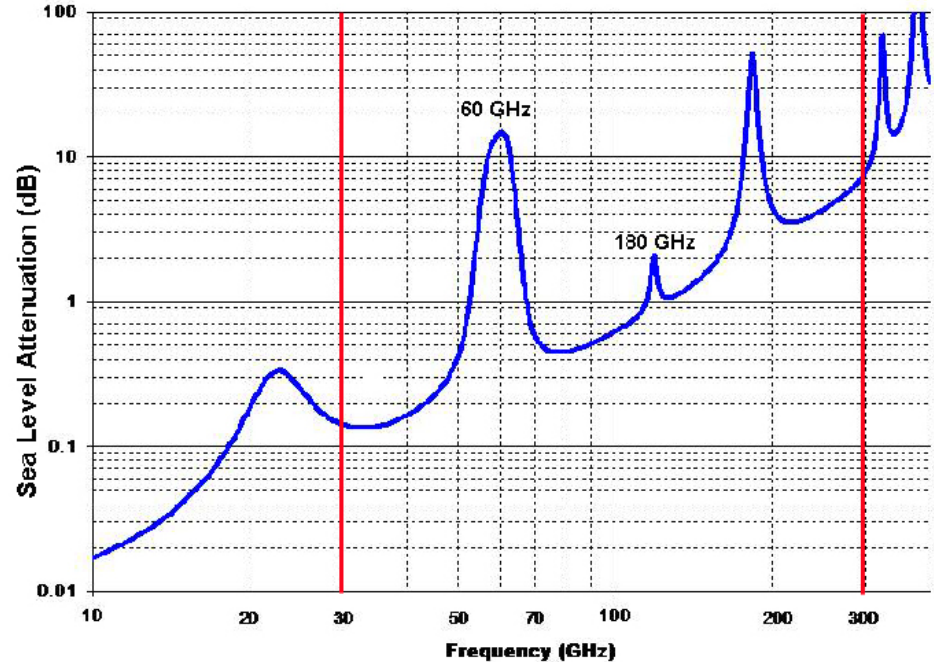
Chart of the Electromagnetic Spectrum



$$\lambda = 3 \times 10^8 / \text{freq} = 1 / (\text{wn} \times 100) = 1.24 \times 10^{-6} / \text{eV}$$

Différents Affaiblissements

Affaiblissement par le Trajet
Affaiblissement par absorption
Affaiblissement dans l'atmosphère
Affaiblissement due à l'eau
Affaiblissement par diffraction
Evanouissement = fading
(multi-trajet)



Affaiblissement

$$\text{LOS } A = 32 + 20 \text{ Log}(F_{\text{MHZ}}) + 20 \text{ Log}(d_{\text{km}})$$

$$\text{NLOS } A = 32 + a \cdot 20 \text{ Log}(F_{\text{MHZ}}) + b \cdot 20 \text{ Log}(d_{\text{km}})$$

$$1 < a < 2 \text{ et } 1 < b < 3$$

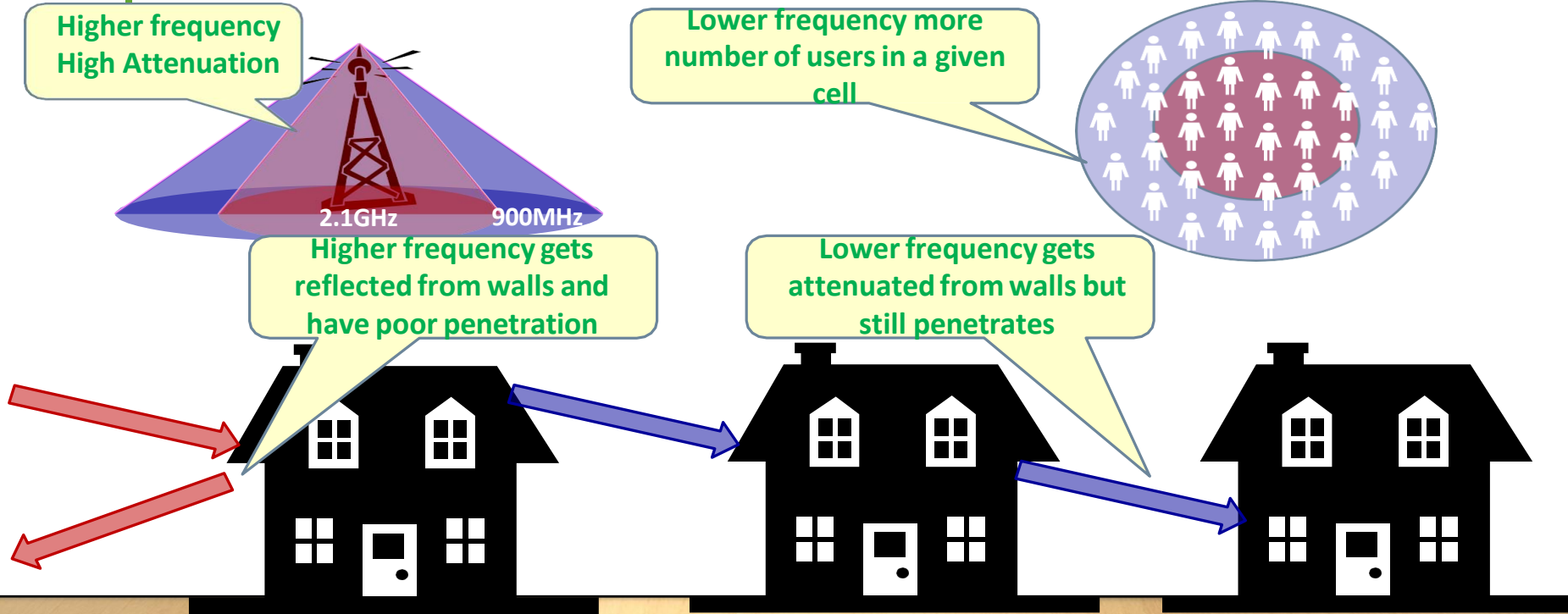
Puissance de 1W (30dBm) à 1mw (0dBm),

$P_{\text{station}} = 10\text{W}$ à 300W

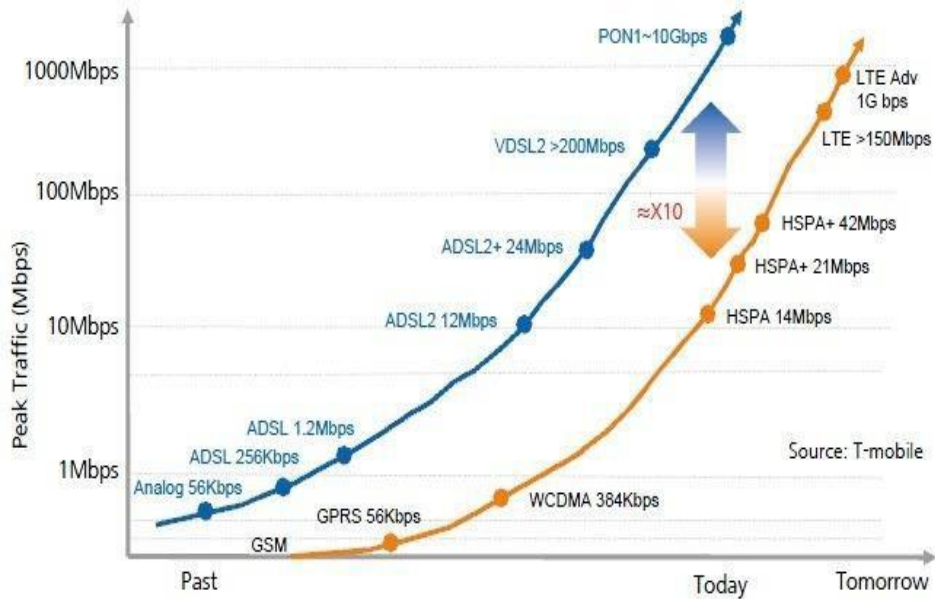
Batterie 20Ah à 40 Ah

Gain d'antenne 3 dBi; 20 dBi

Frequency selection

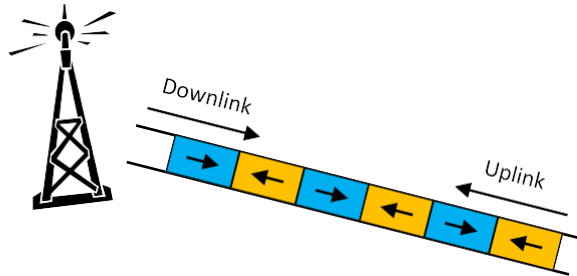
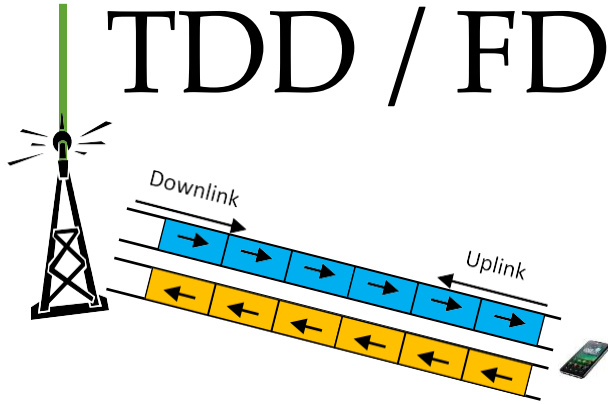


Mobile/Wi-Fi



IEEE Standard	Year Adopted	Frequency	Max. Data Rate	Max. Range
802.11a	1999	5 GHz	54 Mbps	400 ft.
802.11b	1999	2.4 GHz	11 Mbps	450 ft.
802.11g	2003	2.4 GHz	54 Mbps	450 ft.
802.11n	2009	2.4/5 GHz	600 Mbps	825 ft.
802.11ac	2014	5 GHz	1 Gbps	1,000 ft.
802.11ac Wave 2	2015	5 GHz	3.47 Gbps	10 m.
802.11ad	2016	60 GHz	7 Gbps	30 ft.
802.11af	2014	2.4/5 GHz	26.7 Mbps – 568.9 Mbps (depending on channel)	1,000 m.
802.11ah	2016	2.4/5 GHz	347 Mbps	1,000 m.
802.11ax	2019	2.4/5 GHz	10 Gbps	1,000 ft.
802.11ay	late 2019	60 GHz	100 Gbps	300-500 m.
802.11az	2021	60 GHz	Device tracking refresh rate 0.1-0.5 Hz	Accuracy <1m to <0.1m

TDD / FDD



Frequency Division Duplex (FDD)

Simpler to implement

Simultaneous downlink and uplink trans

No need for synchronisation hence simp

Needs paired spectrum UL/DL ratio is fix

Time Division Duplex (TDD)

Implementation is complex

Only uplink (UL) or downlink (DL) at any

Need for synchronisation within the who

No need for paired spectrum

Number of UL/DL ratio is changeable

Latency & Jitter -délai et gigue

Latency is generally defined as the time it takes for a source to send a packet of data to a receiver. In simple terms, half of Ping time. This is also referred to as **one way latency**.

Sometimes the term **Round trip latency** or round trip time (RTT) is also used to define latency. This is the same as ping time.

Jitter is defined as the variation in the delay (or latency) of received packets. 'delay jitter'.

Low latency

End-to-end (E2E) Latency : the time that takes to transfer a given piece of information from a source to a destination, measured at the communication interface, from the moment it is transmitted by the source to the moment it is successfully received at the destination.

1G: Latence moins pertinente

2G : 300-600ms

3G : 100-500 ms

4G : 50-100 ms LTE-A 20ms

5G : 1-10ms

NFV: Moving Hardware into Software

1994

2014

- Virtualisation et SDN



Evolutions

1G : Analogique

2G : Numérique

3G : Data

4G : Débit

5G : Latence

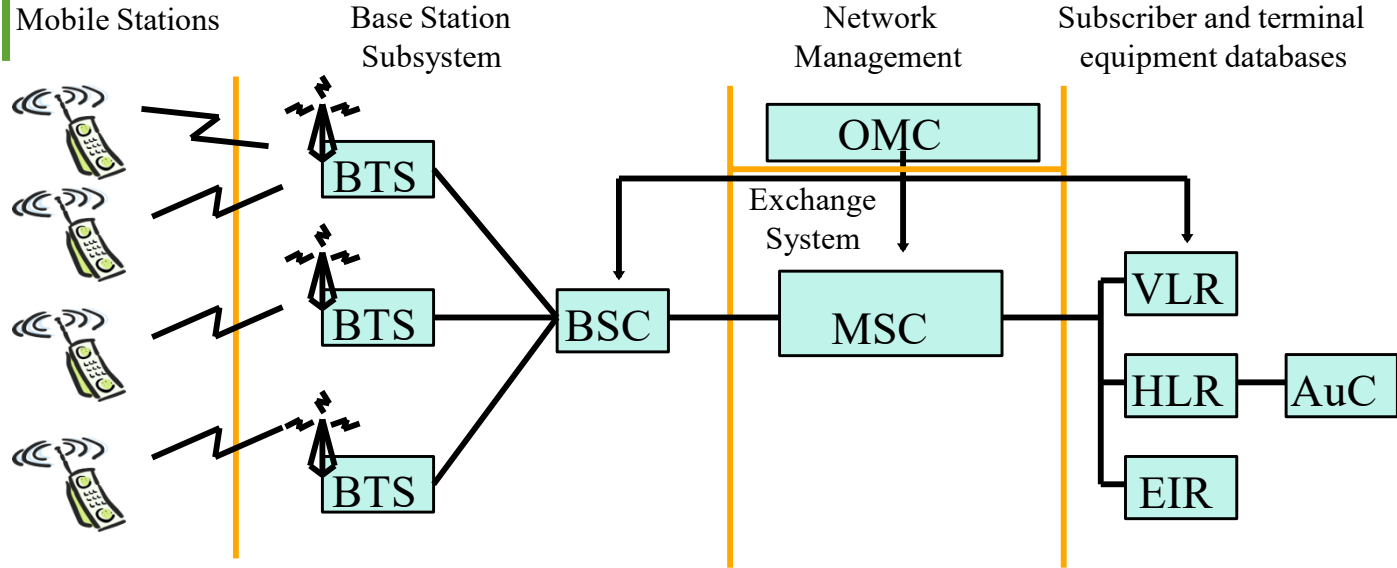
6G : Haute fréquence

7G :

5G Evolution

	Voice Call / SMS	Mobile Data
5G = Fifth Generation	Packet Switched (PS)	
4G = Fourth Generation	Packet Switched (PS)	
3G = Third Generation	Circuit Switched (CS)	Packet Switched (PS)
2G = Second Generation	Circuit Switched (CS)	Packet Switched (PS)
1G = First Generation	Circuit Switched (CS)	N/A

GSM Architecture



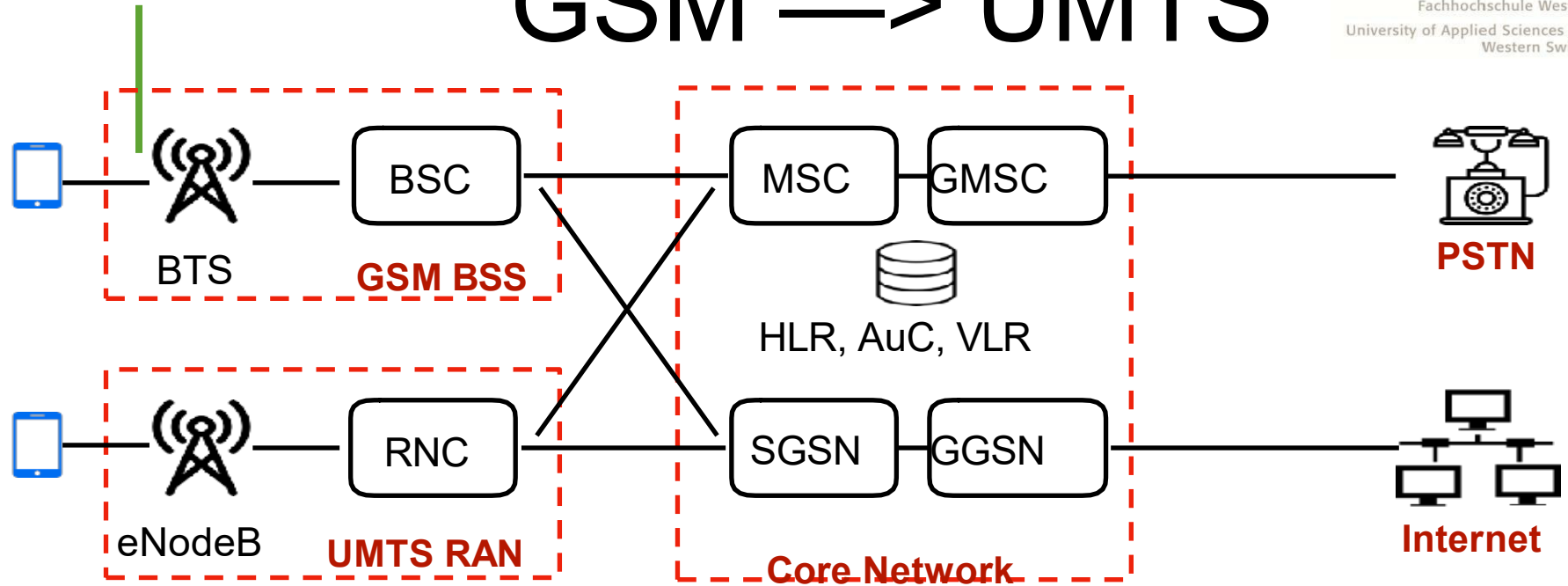
HLR = Home Location Register
AC = Authentication center

VLR = Visitor Location Register
EIR = Equipment Identity Register

GSM Mobile

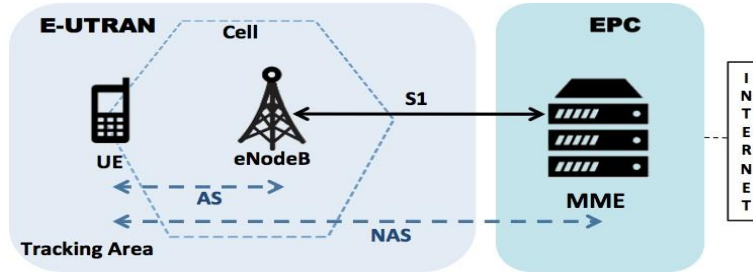
- Mobile Equipment (ME)
- Physical mobile device Identifiers
 - IMEI – International Mobile Equipment Identity
 - Subscriber Identity Module (SIM)
 - Smart Card containing keys, identifiers and algorithms
- Identifiers
 - Ki – Subscriber Authentication Key
 - IMSI – International Mobile Subscriber Identity
 - TMSI – Temporary Mobile Subscriber Identity
 - MSISDN – Mobile Station International Service Digital Network
 - PIN – Personal Identity Number protecting a SIM
 - LAI – location area identity

GSM → UMTS

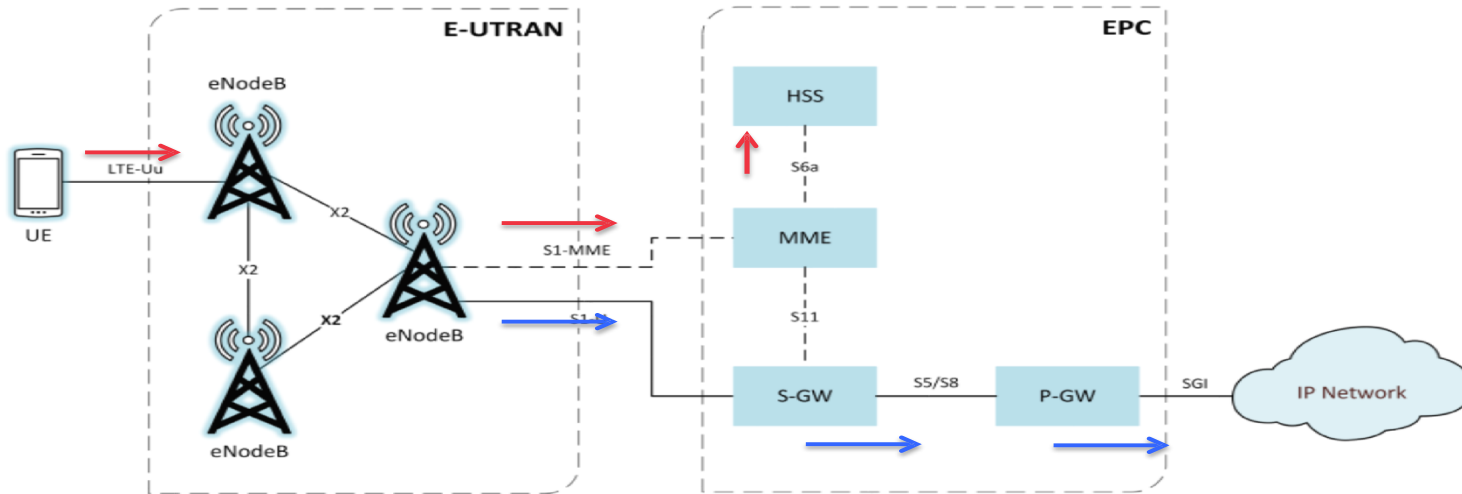


- SIM → USIM
- BTS → eNodeB
- BSC → RNC (Radio Network Controller)
- Backend (MSC, GMSC, SGSN, GGSN all remain the same)

LTE System Architecture



- eNodeB: RF component of LTE
- MME: Mobility Management Entity (Primary Signalling Node)
- HSS: Home subscriber service. Similar to HLR in 2G



Characteristics of LTE Security

- Re-use of UMTS Authentication and Key Agreement (AKA)
- Extended Key Hierarchy
- Possibility for longer keys
- Greater backhaul protection

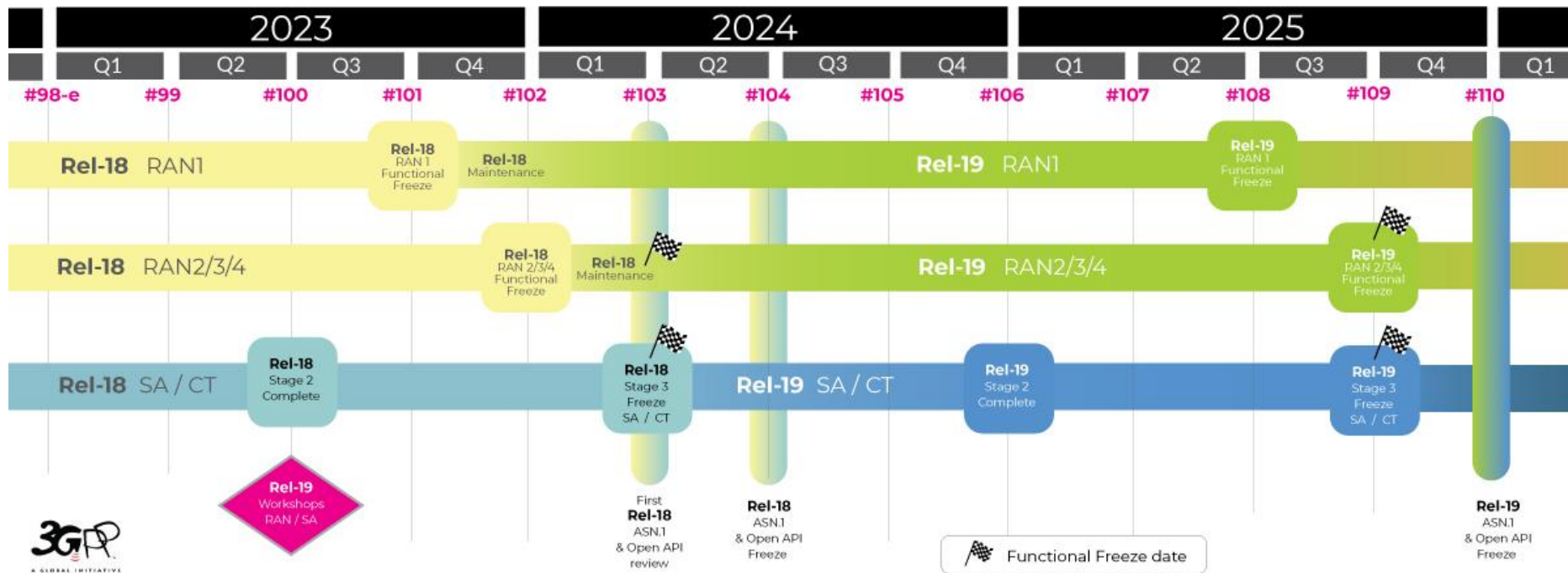
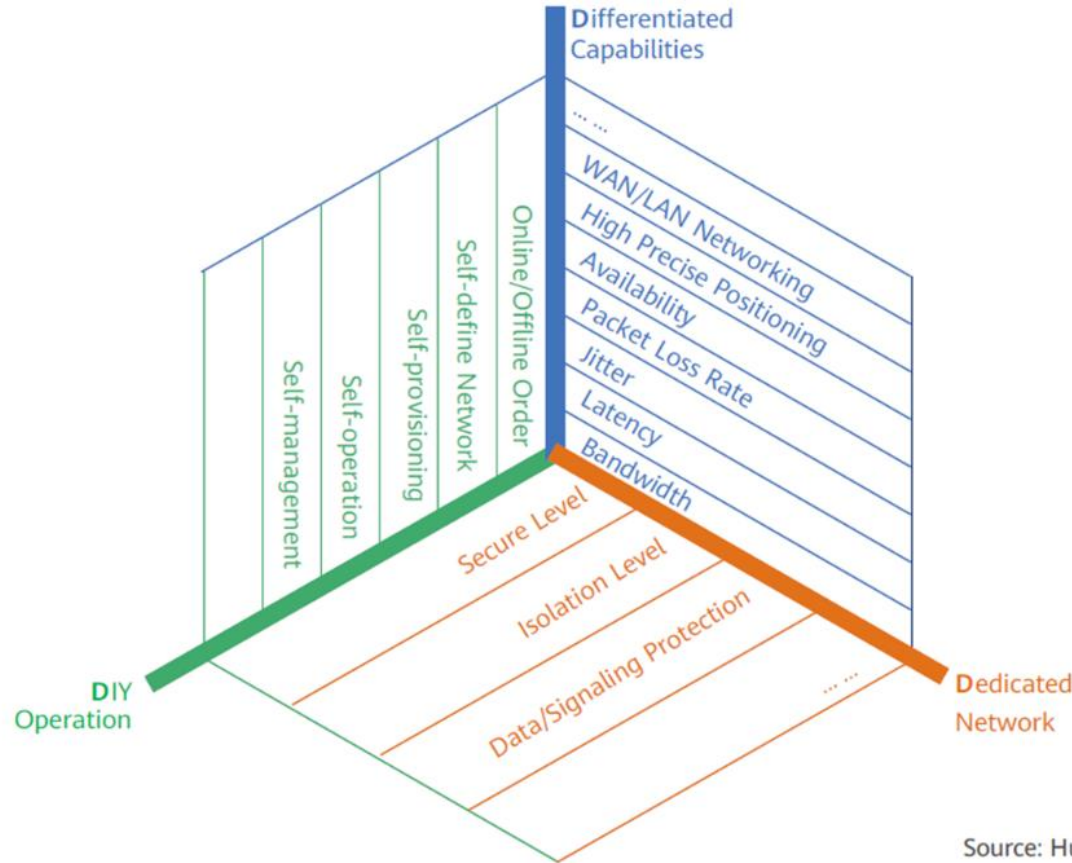


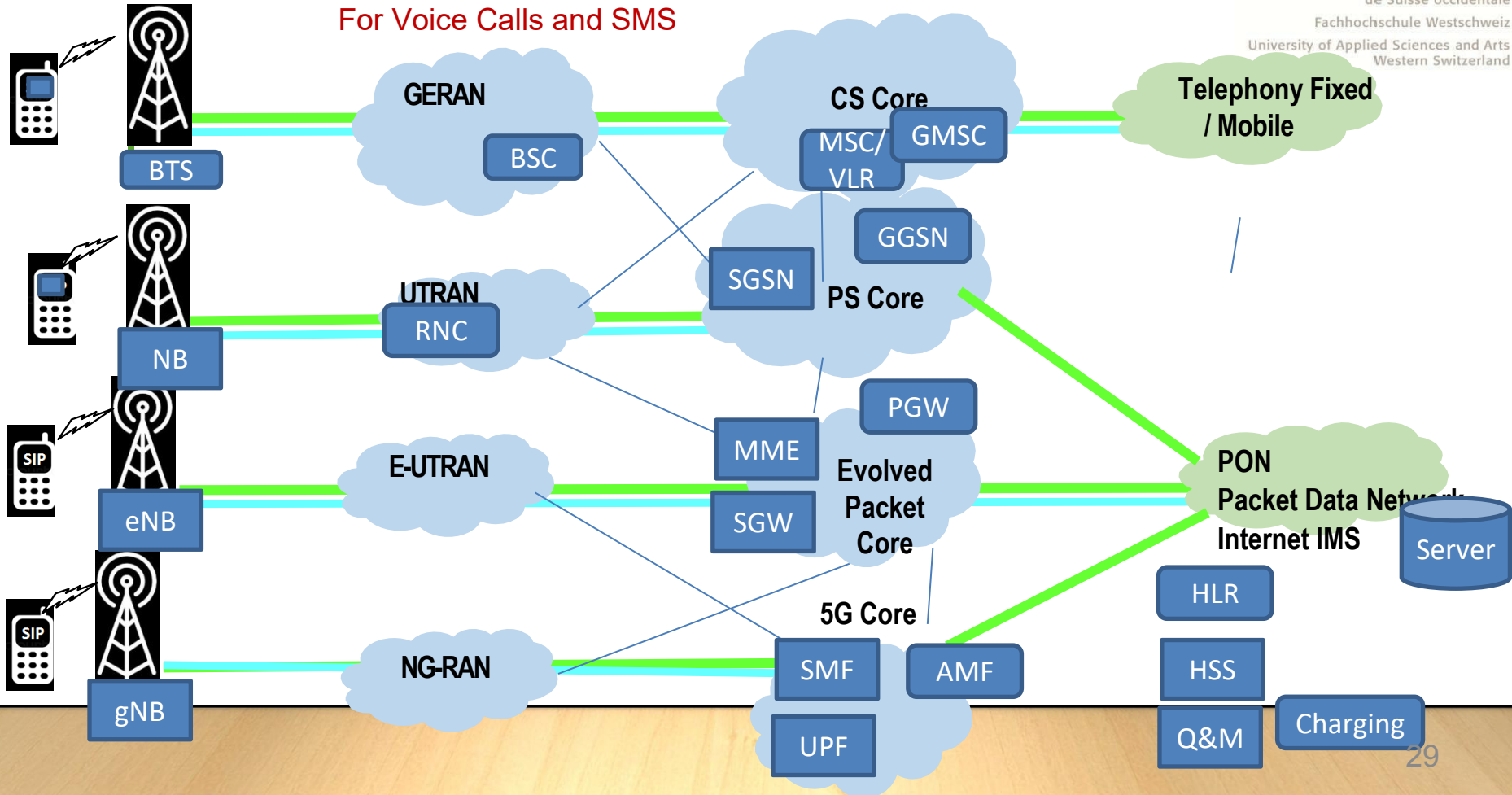
Figure 9: 3D Dimensions of Network SLA for Industry Digital Transformation

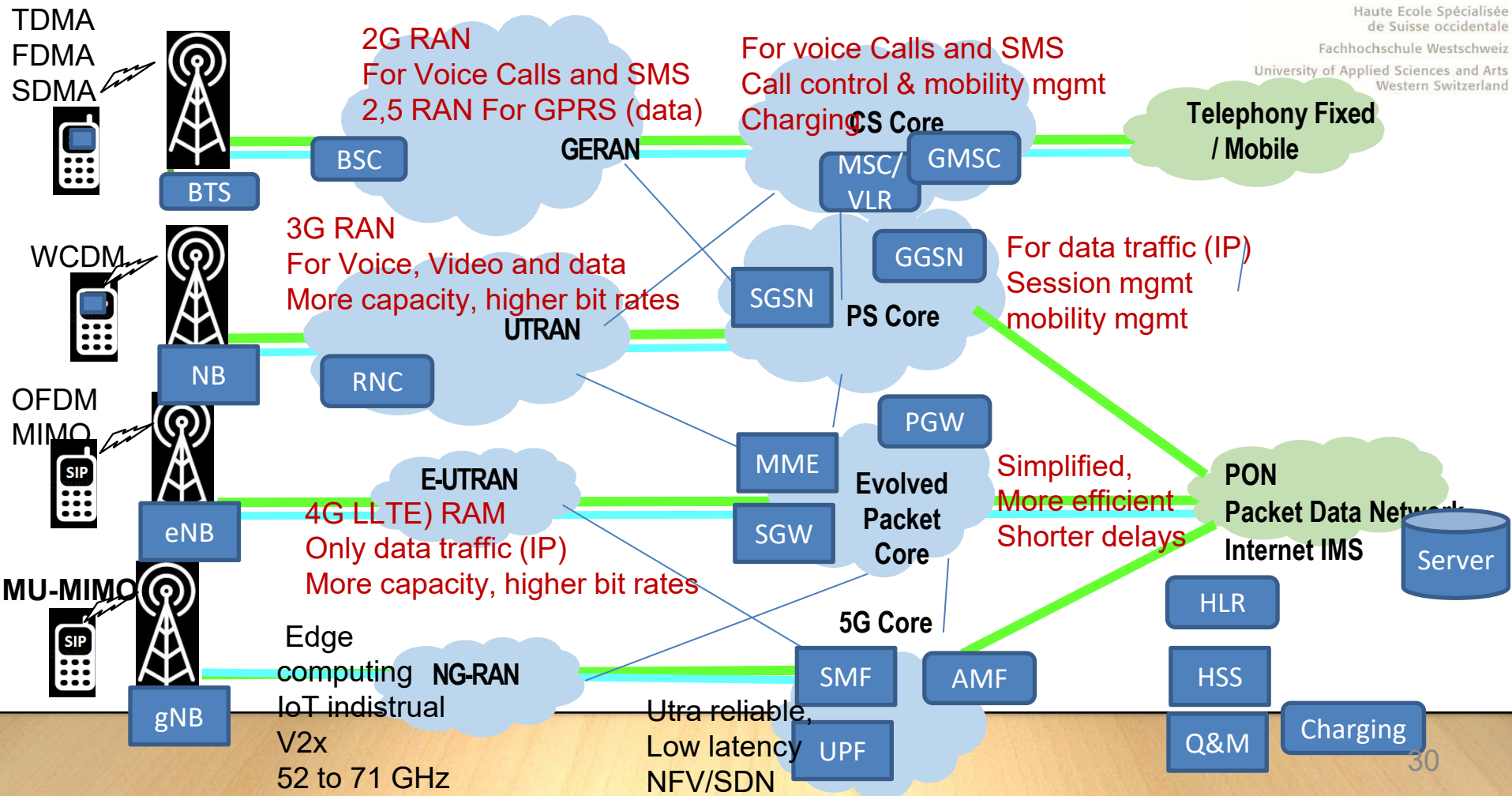


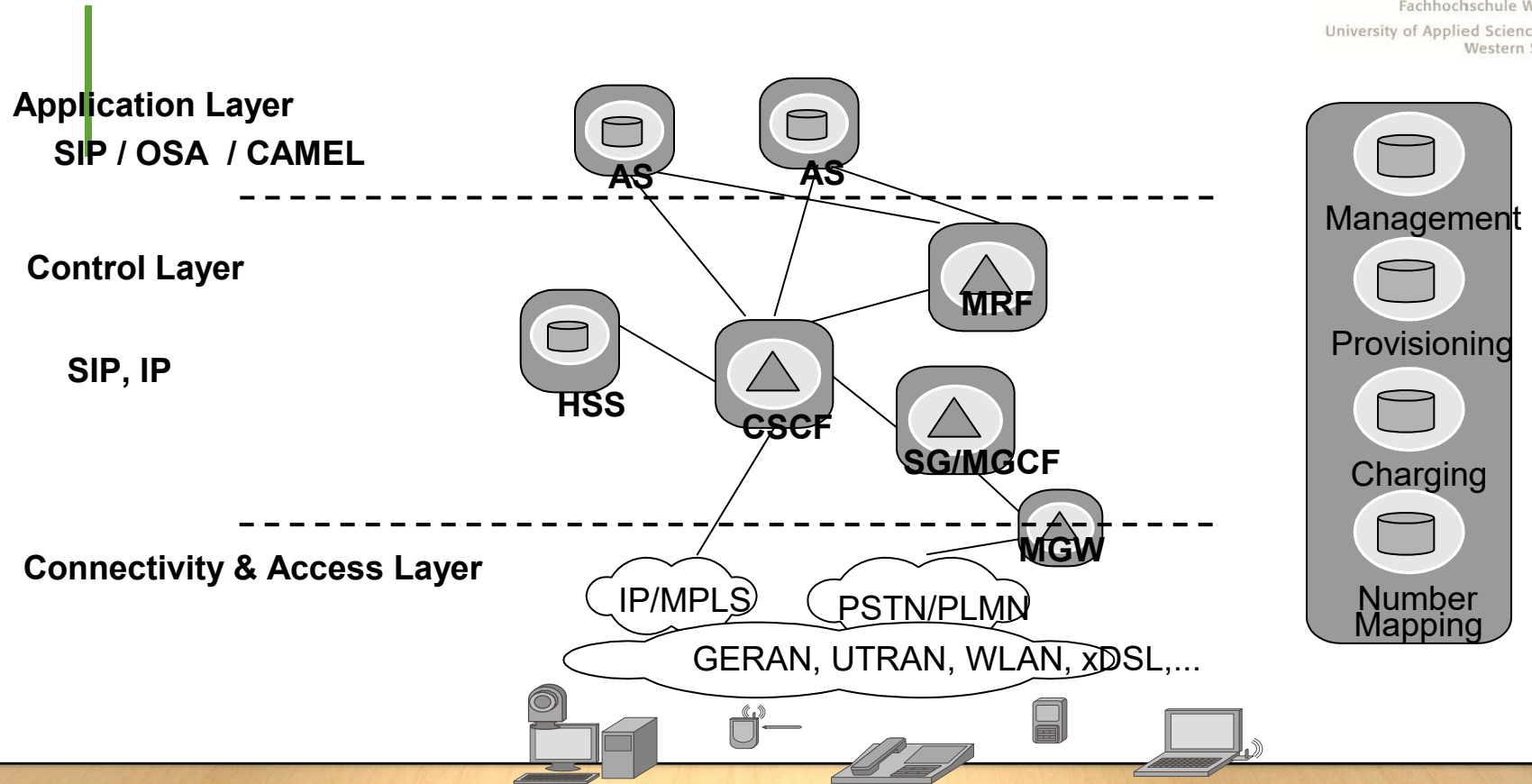
Source: Huawei Clab & CAICT

Ex.
For Voice Calls and SMS

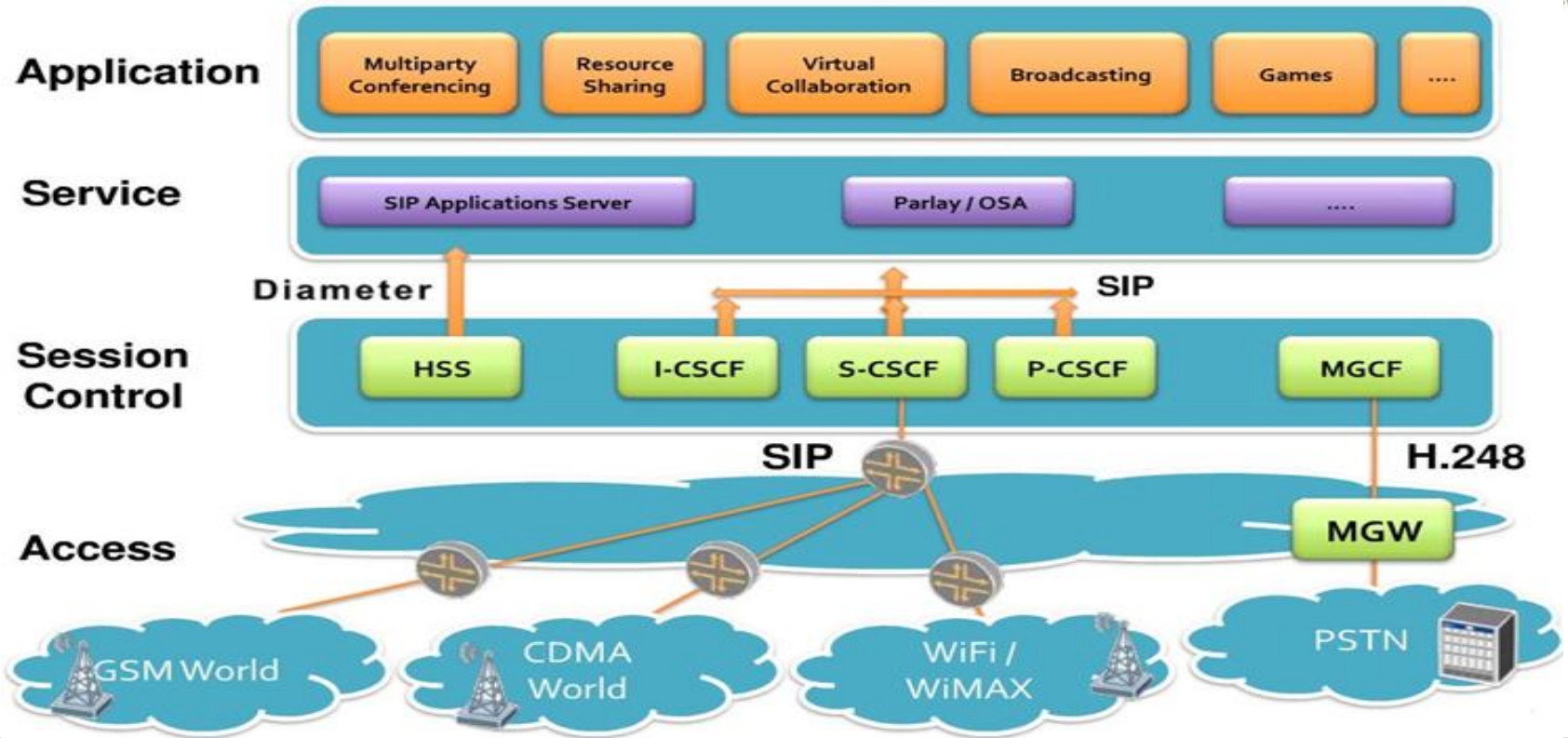
3 functionalities for each network



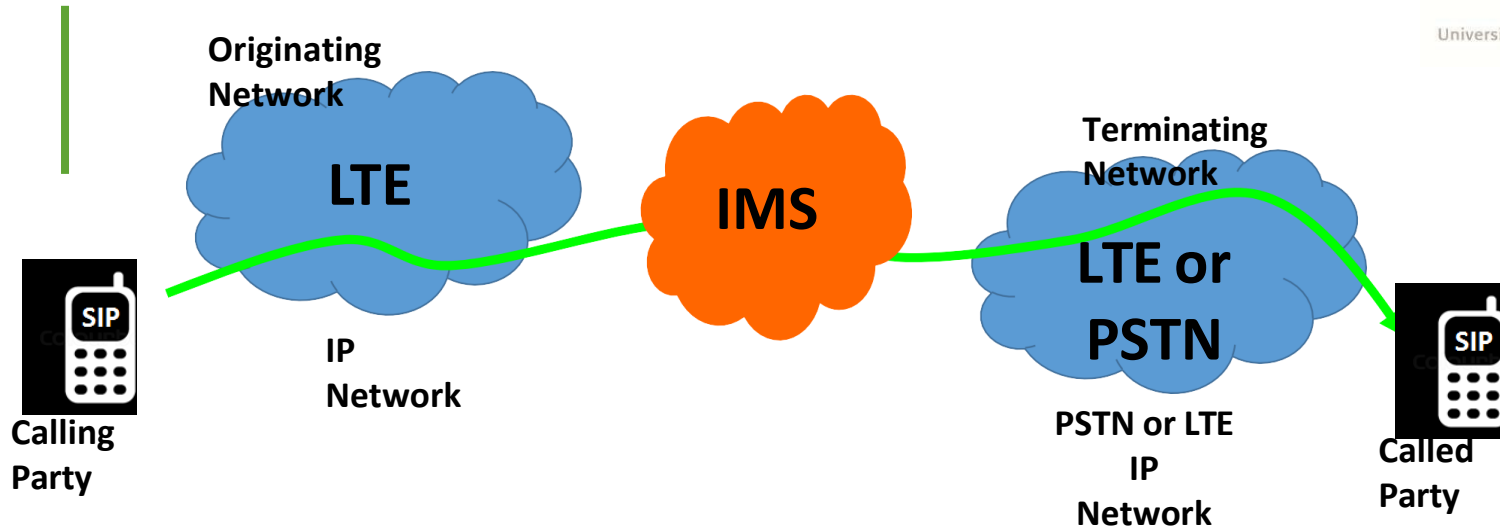




Architecture réseau IMS



What is VoLTE



IMS enables a packet-based network to provide multiple services on single Control/Service Layers via different access network.

Key Drivers to IMS

- Access agnostic
- Services independent
- Open architecture
- Multi-device
- Vendor independent

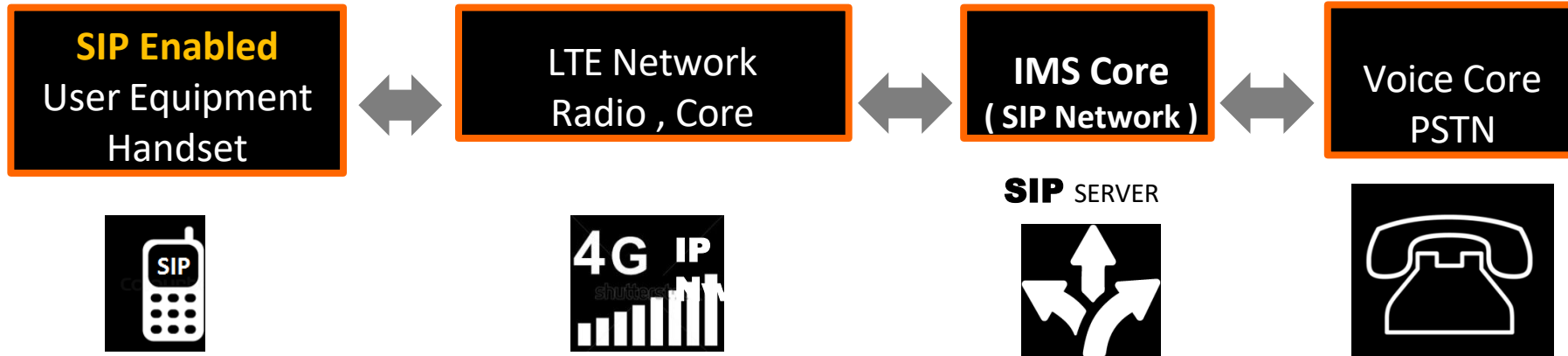
VoLTE Building blocks

VoLTE : Voice over LTE

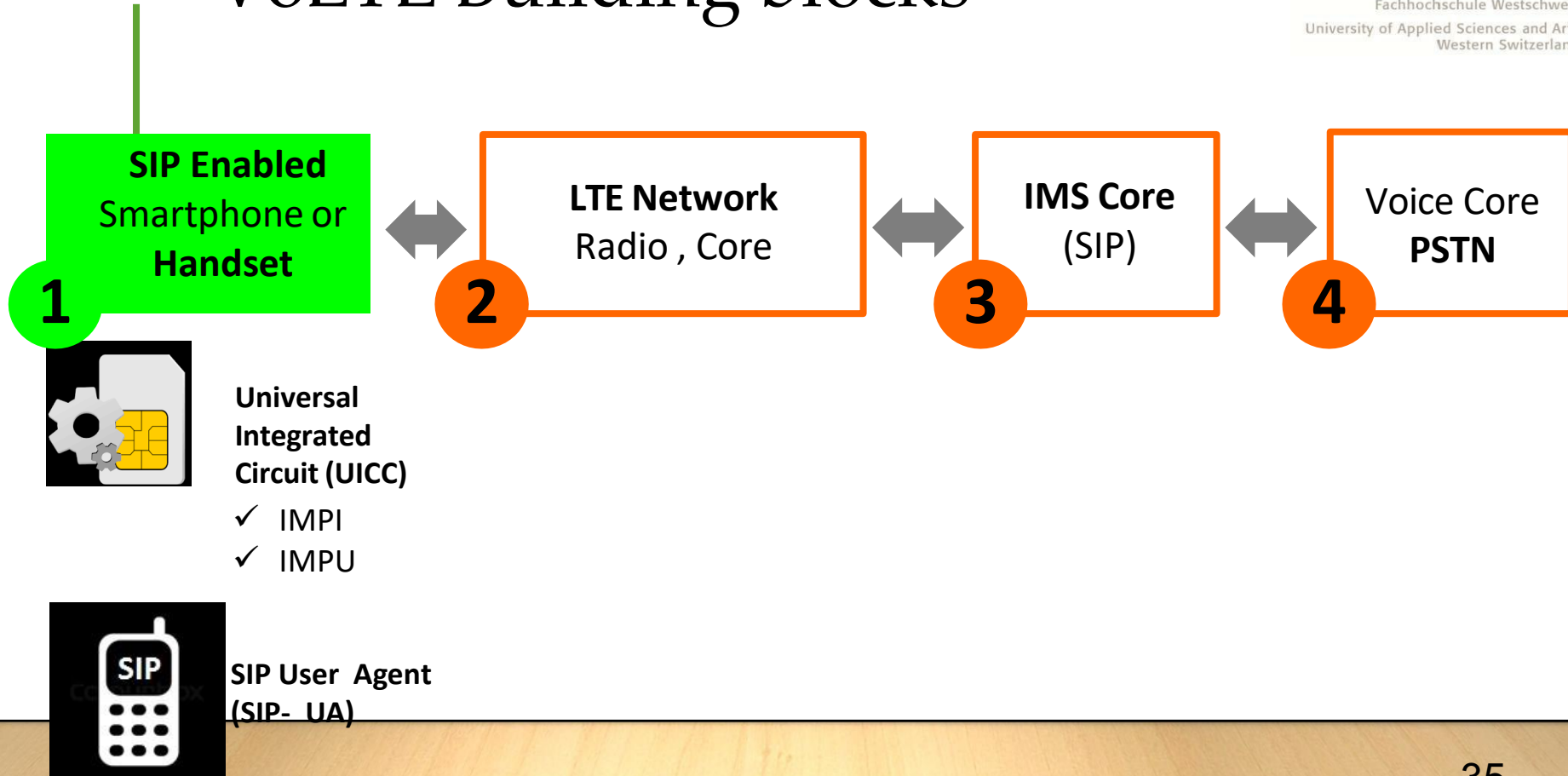
- Voice calls over a 4G LTE network
- LTE Packet Network used for carrying voice payload over all IP Network

VoLTE Benefits to users

- HD Voice
- Fast Call connection (0.25s)

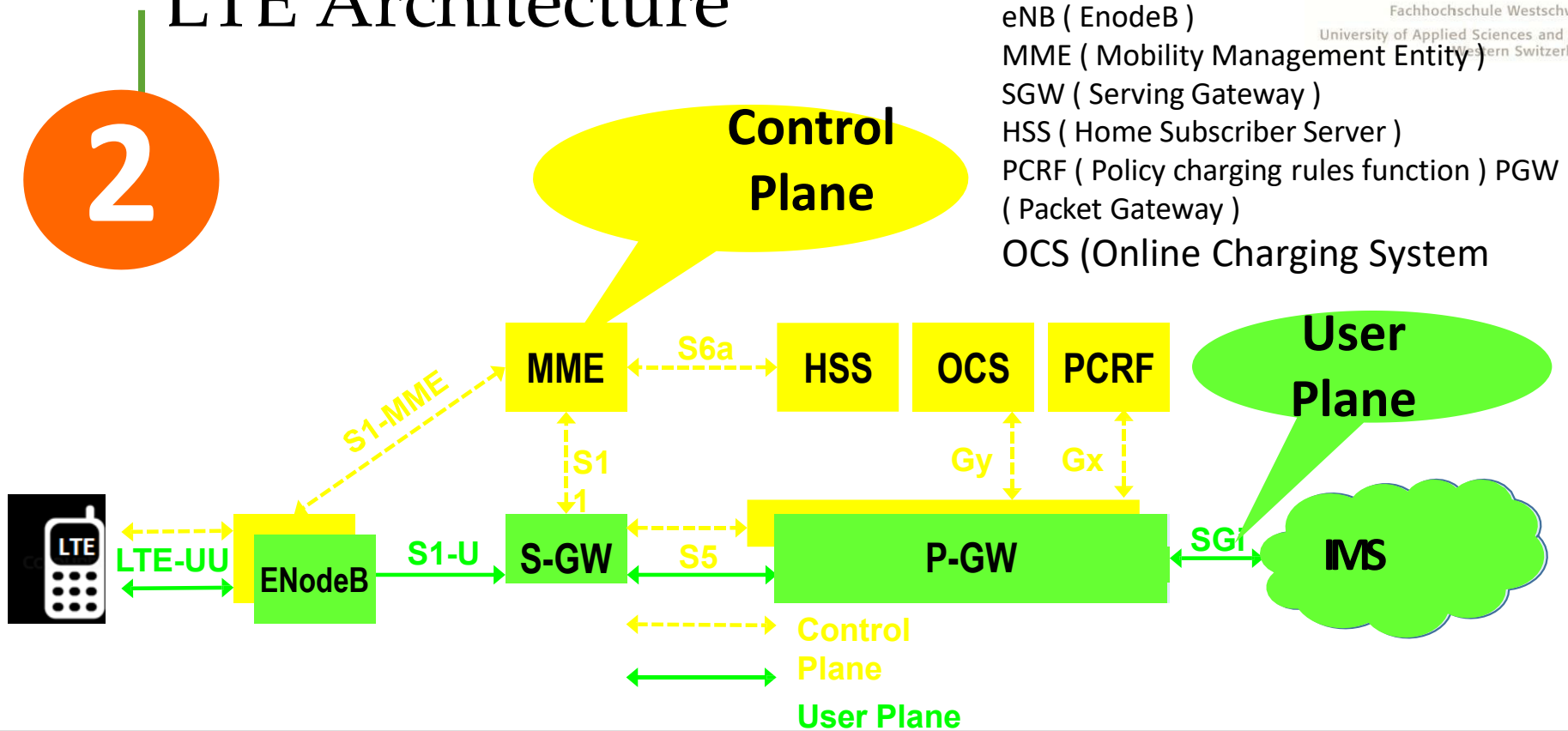


VoLTE Building blocks



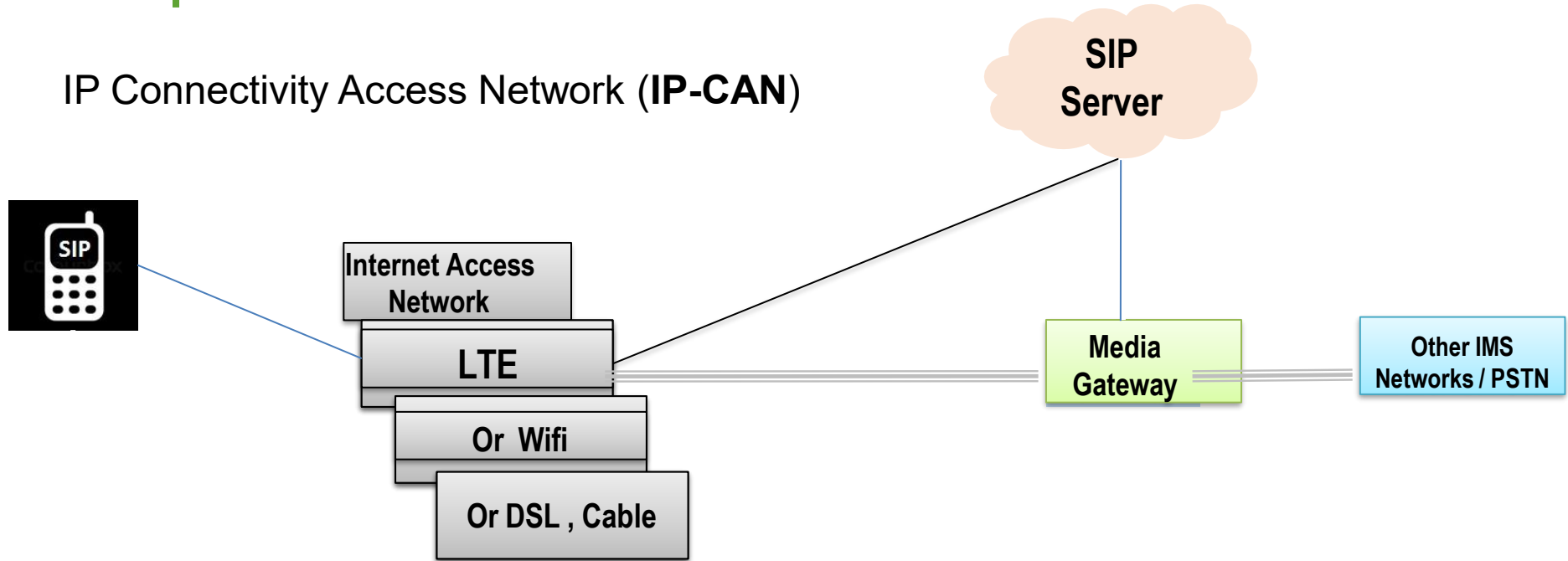
LTE Architecture

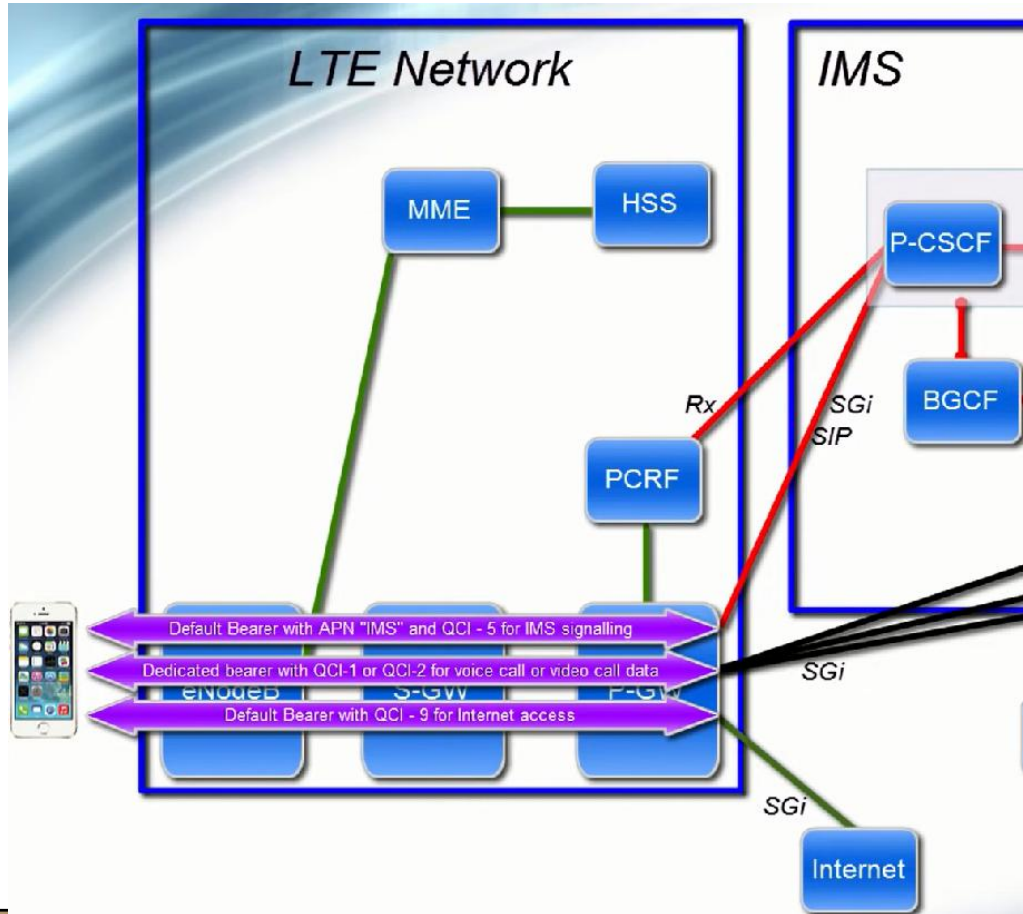
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Role of IPCAN

IP Connectivity Access Network (IP-CAN)





What if LTE Radio is Congested ??

Priority	QCI	Resource Type	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	5	non-GBR	100ms	10^{-6}	Volte , IMS Signaling
2	1	Guaranteed Bit Rate (GBR)	100ms	10^{-2}	Volte , Conversational Voice
3	3		50ms	10^{-3}	Real Time Gaming, V2X messages
4	2		150ms	10^{-3}	Volte , Conversational Video (Live Streaming)
5	4		300ms	10^{-6}	Non-Conversational Video (Buffered Streaming)
6	6	non-GBR Best Effort	300ms	10^{-6}	Internet Services , Video (Buffered Streaming) TCP-Based (for example, www, email, chat, ftp, p2p and the like)
7	7		100ms	10^{-3}	
8	8		300ms	10^{-6}	
9	9		300ms	10^{-6}	

Changes in Core Network to support VoLTE

- Default & dedicated Bearer Enable
- IMS APN IP Pool
- PGW - SBC / PCSCF Reachability
- Disable Gy for IMS

**Packet Core
SGW / PGW**



- Sv Link – SRVCC
- IMS APN SGW / PGW Selection
- Paging Policy - VoLTE Calls
- QCI – 1,5 Testing & Validation

**Packet Core
MME**



- Sv Link – SRVCC
- IMS & MGW – IP Reachability
- Break in & Break out Calls Config
- Common Codec between MSS & IMS

**Voice Core
MSS / MSC**

References

- GSMA FCM.01
- GSMA IR.92 (Résumé)
- IMS SPECS 3GPP 23.228